COMP 581: Introduction to Robotics

Fall 2026
Instructor: Prof. Ron Alterovitz
Teaching Assistant: Kamal Deep Vasireddy <kamalv@unc.edu>

Lab 1: MicroPython Overview and Your First Robot

Deadlines:

October 20, 2026, at the beginning of class. For the late policy, see the course syllabus. **Before the due time, each team should submit their team's code in Gradescope**; see specific instructions in the Code section below**.**

Submission Deadlines:

Optional Pre-Lab Reflection:	10/16 at 11:59pm
Submit Code and Run Lab:	10/20 at 9:30am
Optional Post-Lab Reflection:	10/23 at 11:59pm

Lab Location: FB241A

The Objective

The primary purpose of this lab is to get some experience with Lego Mindstorms EV3 and the Ev3 MicroPython API. You will build your first mobile robot and familiarize yourselves with the motors and sensors.

Task 1: Read our EV3 MicroPython Tutorial

If you have not already set up the Lego Mindstorms EV3 with your laptop, you can do so now. It is important that you skip all sections involving the SD card, ours are already configured. The sections we will need for this class are as follows:

1.1, 1.2, 1.5.1, 1.5.2, 2.1-End

Start by reading the Tutorial at:
https://education.lego.com/v3/assets/blt293eea581807678a/bltb470b9ea6e38f8d4/5f8802fc4376310c19e33714/getting-started-with-micropython-v2_enus.pdf

For more information about EV3 MicroPython, see:
https://pybricks.com/ev3-micropython/index.html

Task 2: Build Version 1 of Your Robot

You should build a robot to accomplish the objectives in Task 3. The robot should be designed to operate on a flat floor with low-pile carpet.

Your robot must be built only using pieces from your Lego Mindstorms kit. You should design your robot such that the dark gray center button may be pressed without moving the robot and such that pushing the button does not disrupt any of the sensors. Your robot should remain in one piece at all times, its diameter with respect to the ground should not exceed 40 cm, and its diameter shall not change substantially during the task. All measurements will be made with respect to the *measuring point*: a distinct point of your choosing that is on the **frontmost surface** of your robot. More specifically, your robot will be placed on the ground behind a starting line drawn on the ground, and your chosen measuring point must be a point on your robot directly above the starting line, and no part of your robot is permitted to be in front of the starting line. Your robot should be designed to accomplish the task autonomously without any human intervention (except for pushing the dark gray center button as required).

Task 3: Odometry, Ranging, and Bumping

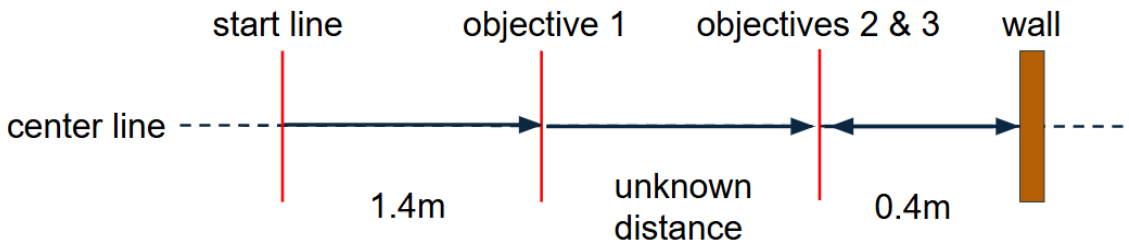
All the objectives in this lab are to be completed autonomously in a single run, without picking up your robot. The only human intervention is to push the center dark gray button. All tasks should be completed within 90 seconds total (the timer will be paused while taking measurements).

Objective 1 (Odometry): You will place your robot behind a starting line such that its measuring point will be on a specific *starting point* on the starting line. You should then press the center dark gray button. Upon pressing the button, your robot should move forward in a straight line for 1.4 meters and stop. Immediately after stopping, the robot should beep. Here, we will measure the distance from the starting line and deviation from the center line (movement left and right), and points will be awarded based on accuracy. To enable time for measurement, the robot should remain stopped until the button is pushed.

Objective 2 (Ranging): After completing the first objective, you will press the center dark gray button, and your robot should resume moving forward. Somewhere directly in front of your robot will be a flat surface (like a wall or large object) that you can assume is at least 17 cm high and can be detected by the ultrasonic sensor. This wall will lie within 2.5 meters of the stopping point defined in Objective 1. Therefore, the total distance between the start line and the wall will be less than 3.9m. Note that this is simply an upper bound; the unknown distance may be any length such that this upper bound holds. The robot should stop such that it is 40 cm from the flat surface (measured from the nearest point on the robot). Your robot should beep and pause until the center dark gray button is pressed.

Objective 3 (Bumping): After the previous two objectives, you will push the center dark gray button, and your robot should now move forward and approach the surface. It should bump the surface with a bump sensor placed on the front of the robot. After bumping into the surface, the robot should move in reverse until it is 40 cm from the wall again.

A visual representation of the workspace looking down from the ceiling:



Points

Logistics:
Turn in code on time: 10 points.
Performance:
Attempt: 25 points. You will receive these points simply by having EV3 MicroPython successfully installed, showing up to the lab on time, and having your robot attempt the course.
Odometry: 20 points. 10 points will be awarded based on distance traveled and 10 points for minimizing deviation.
Ranging: 25 points. 15 points will be awarded for distance from wall and 10 points for deviation from the center line.
Bump sensing: 20 points. 10 points will be awarded for bumping the wall and reversing, 5 for achieving the correct distance from the wall, and 5 for deviation from the center line.

Code

To complete this lab you will use a subset of the EV3 MicroPython API:
https://education.lego.com/v3/assets/blt293eea581807678a/bltb470b9ea6e38f8d4/5f8802fc4376310c19e33714/getting-started-with-micropython-v2_enus.pdf
Important: You may only use the following modules from EV3 MicroPython API (covered in chapters 3-6 of the API document above):
ev3brick
ev3devices
parameters
tools
You may *not* import, use, or copy the “robotics” module, or any of its included classes or functions, in your program.

A robot relying on a module not allowed as stated above will be disqualified. Also, any robot using Wi-Fi or Bluetooth communications in any way will be disqualified.

You must submit on **Gradescope** a *zip* file of your Python source code files. The *zip* file should include all the Python source code files you wrote (which should have a *.py* file extension). In the comments at the top of the Python source file containing your *main.py* file, please include

the **names** and **PIDs** of all team members

Optional: Lab Reflections

Optional reflections will award points towards the performance portion of the lab and are designed to prompt students to think systematically and critically about the tasks. You must have attempted a lab in order to receive credit for a post-lab reflection. These should be submitted as a *pdf* on **Gradescope** for each team. Each reflection you complete is worth 20 points and will scale the performance part of your lab score (i.e., the 90 points excluding turning in your code) to be out of 70 or 50 points. For example, if you do both reflections, turn in your code on time, and get 70/90 for robot performance, your score will be 10 (for turning in code) + 40 (for doing both reflections) + 50*(70/90) = 89 points.

Pre-Lab Reflection

Task 1:

Write a short description of your robot design in terms of the components used and how they help accomplish the task. Examples include motors, sensors, locomotion type (wheels, tracks, legs, etc…), or any other relevant components.

Task 2:

Draw a high-level flowchart of the robot's functionality.

Task 3:

Write the pseudocode that implements your flowchart.

Post-Lab Reflection

Task 1:

Write a short description of your robot's final design and how it differs from your initial concept. Why did you choose to make those changes?

Task 2:

Consider how your robot performed in the lab and how it was different from your expectation. Were there any particular shortcomings that you identified?

Task 3:

Explain what changes you would make to your design or implementation to achieve better results for the particular task.

Collaboration

You should feel free to discuss concepts in the course with other students in natural language (e.g. English). You should not share any programming code with others and must write all the robot's programming code yourself. If you access any sources other than the textbook or documents on Canvas, you must cite them in comments at the top of your code and send an email to the instructor with the citation. You must fully understand your code and be able to reproduce the algorithmic approach without references if asked. The Honor Code is in effect for these policies.

Good luck!